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Original Research

Unraveling the Spectrum of Ocular Toxicity with Oxaliplatin: Clinical Feature Analysis of Cases and Pharmacovigilance Assessment of the US Food and Drug Administration Adverse Event Reporting System Database

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ABSTRACT

Purpose: Ocular adverse events (oAEs) are a class of adverse events associated with oxaliplatin that are realistically observed in real-world settings. Herein, we aim to describe the clinical characteristics of oAEs associated with oxaliplatin through a systematic review of case reports and to assess a potential safety signal.
Methods: PubMed, Embase, and Cochrane Library databases were used to retrieve case reports. The global disproportionality study was performed leveraging the US Food and Drug Administration Adverse Event Reporting System database from January 2004 to September 2023. Bayesian information component (IC) and reporting odds ratio (ROR) were applied to identify and evaluate potential oAEs associated oxaliplatin.
Findings: A total of 20 cases from the systematic case review (of 13 screened articles) were reported on oAEs associated with oxaliplatin, with ages between 26 and 76 years. Therein, 16 (84.2%) cases described loss of vision, and the remaining cases presented with bilateral blepharoptosis, papilledema, and optic disc swelling. Insights from the US Food and Drug Administration Adverse Event Reporting System database showed that oAEs accounted for 4.28% (n = 1194) of the overall oxaliplatin-related adverse event reports, of which 1140 (95.48%) had a serious outcome. The median (interquartile range) onset time of oAEs with oxaliplatin was day 1 (0–25; n = 649). Disproportionality analysis revealed that ocular injuries NEC (n = 28, ROR, 22.72; lower limit of the 95% 2-sided CI for IC, 3.12) was the most significant signals detected. Additionally, unexpected significant oAEs, including eyelid ptosis, eyelid edema, eye movement disorder, blepharospasm, periorbital edema, swelling of eyelid, ophthalmoplegia, retinal vein thrombosis, cataract nuclear, blindness cortical, cataract subcapsular, and lacrimation disorder, were also reported disproportionality.
Implications: Our study systematically described the characteristics and outcomes of oxaliplatin-related ocular toxicity and also uncovered potential oAEs that were not disclosed in the package insert. Further prospective epidemiologic studies to validate these findings are warranted.

Introduction

Oxaliplatin, a third-generation platinum derivative, exerts its antitumor effect by forming DNA adducts that interfere with DNA replication and transcription in tumor cells.[1] Nowadays, oxaliplatin combined with fluorouracil and leucovorin (FOLFOX) or with capecitabine has emerged as the standard regimen of treatment for colorectal cancer in adjuvant or metastatic settings[2] and has also been shown to improve clinical outcomes in ovarian cancer, lung cancer, pancreatic cancer, and gastric cancer.[3–6] Nonetheless, their collateral effect still poses a great challenge to the clinical application of oxaliplatin.[7]

The major adverse events (AEs) of oxaliplatin are peripheral neurotoxicity, gastrointestinal toxicity (diarrhea, vomiting, and nausea), and mild hematologic toxicity (thrombocytopenia, neutropenia, and anemia).[8,9] In particular, oxaliplatin has also been documented to be associated with a broad spectrum of ocular AEs (oAEs) including visual loss, blurred vision, tunnel vision, and altered color vision and may induce ocular nerve changes.[10,11] Oxaliplatin-related oAEs are uncommon and

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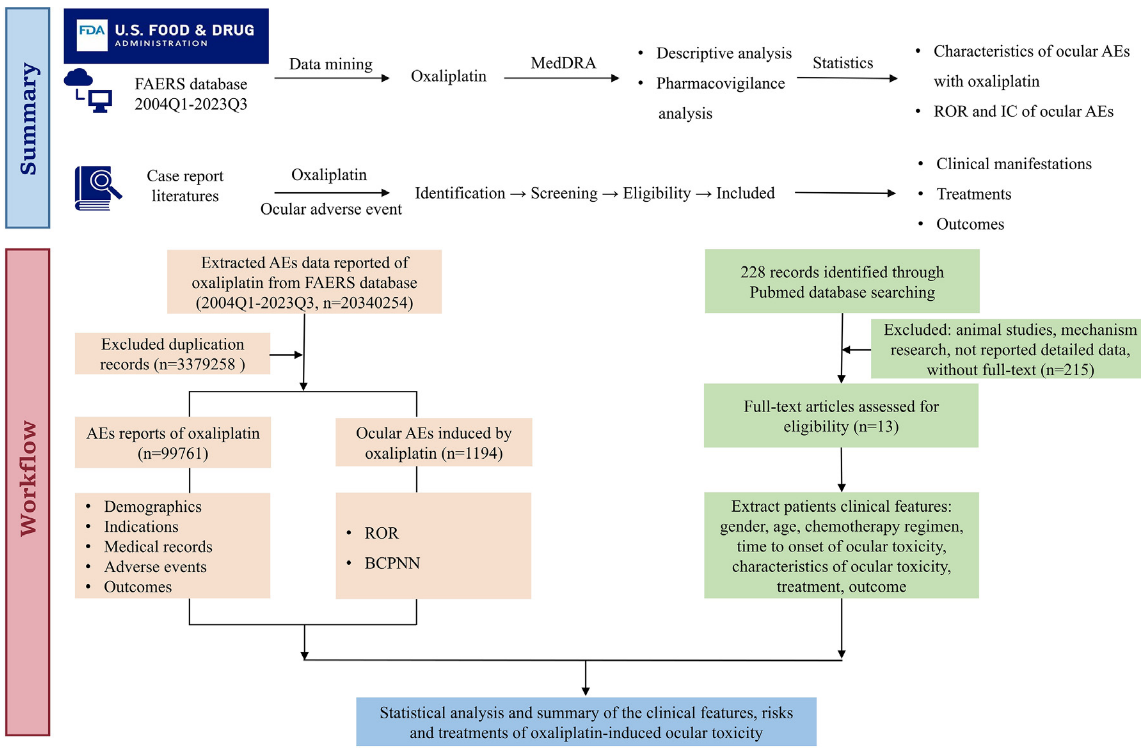


Figure 1. Overview for the ocular events related to the oxaliplatin from US Food and Drug Administration Adverse Event Reporting System (FAERS) database. AE = adverse event; BCPNN = Bayesian Confidence Propagation Neural Network; FDA = US Food and Drug Administration; IC = information component; MedDRA = Medical Dictionary for Regulatory Activities; ROR = reported odds ratio.

rarely investigated in prospective trials but have been reported sporadically in post-marketing case reports and small case series.[1,11,12] In a previous retrospective study,[13] real-world data indicated that approximately 38.61% of 365 patients with metastatic colorectal cancer who were treated with the FOLFOX regimen experienced oAEs. The most frequent oAEs seen were visual loss (58%), tearing (52%), blurred vision (34%), dry eyes (29%), and conjunctivitis (15%). Although none of these patients had permanent damage and the role of fluorouracil in oAEs cannot be excluded, the ocular toxicity of oxaliplatin seems to be underestimated. Meanwhile, in light of the limited sample size, the presented findings provide only a partial epidemiologic perspective. Hence, a systematic summary regarding the clinical manifestations, incidence, onset time, management, and outcomes of oAEs associated oxaliplatin in a real-world setting with a large sample size was highly desirable. The US Food and Drug Administration Adverse Event Reporting System (FAERS) is a global, authoritative, open-source database for the spontaneous reporting of AEs. It contains potential AEs that have occurred in large real-world samples of drugs under conditions that are not addressed by controlled studies.[14]

Therefore, considering these aspects, the aim of this study was to describe the clinical characteristics of the oAEs associated with oxaliplatin, leveraging the review of the case reports and FAERS database, and thereby providing vigilance reference for clinicians or health care professionals to recognize and manage this uncommon AEs.

Materials and Methods

Summary of the Research

First, we performed a retrospective disproportionality analysis leveraging the large sample, real-world database (FAERS) to (1) investigate the association between oAEs and oxaliplatin therapies, (2) describe the clinical characteristics of oAEs induced by oxaliplatin, and (3) uncover the potential oAEs related to oxaliplatin. Second, we conducted a systematic literature review to further confirm whether there is an association between oAEs and oxaliplatin and summarize the clinical manifestations, management, and outcomes of oxaliplatin-related oAEs. The flow chart of this study is displayed in Figure 1.

Pharmacovigilance Analysis

Study Design and Data Source

An observational, retrospective, case/noncase pharmacovigilance study was designed to explore oxaliplatin-related oAEs leveraging the FAERS database. Oxaliplatin[a,b] was used as the defined drug of interest to obtain report data from the FAERS Publish Dashboard for the 2 decades from January 1, 2004, to September 31, 2023 (https://fis.fda.gov/extensions/fpd-qde-faers/fpd-qde-faers.html). Because the patient records in the FAERS databases are publicly available, anonymized, and deidentified, no informed consent or ethical approval was required.

Procedures

The data extraction and mining processes of the present study are illustrated in Figure 1. Oxaliplatin-associated oAE reports were identified as cases by searching the Medical Dictionary for Regulatory Activities (version 26.1), for Preferred Terms (PTs) under the System Organ Class “Eye disorders (System Organ Class: 10015919)”. On the contrary, reports of other AEs were considered noncases. The dataset of patient demographic information (gender and age), reporting characteristics (reporting year, reporting region, and occupation of reporters), and medical information (indications and drugs) were collected for each detected AE report. In the FAERS database, it is possible for a reported AE case (patient) to be associated with multiple AE reports. Thus, a 3-step deduplication procedure was employed in this study in accordance with the

[a] Trademark: ELOXATIN®: Sanofi Pharmaceutical Co., Ltd.
[b] Trademark: ELPLAT®: Yakult Honsha Co., Ltd.

Table 1
Demographic and clinical information of reports with oxaliplatin-related ocular adverse events sourced from the US Food and Drug Administration Adverse Event Reporting System database (from January 1, 2004 to September 31, 2023).

Characteristics	Ocular Events	Other Events	*P* Value
Number of events	1194	98,567	
Gender			
Female	574	39,812	<0.001
Male	539	50,019	
Unknown	81	8736	
Patient's age			
Data available	991	79,550	
Age (y), median (Q1, Q3)	63 (53.5, 71)	63 (55, 70)	
Age group			0.297
<18 y	1	304	
≥18 and <65 y	547	43,015	
≥65 y	443	36,231	
Indications (top 5)	Colorectal cancer, 638	Colorectal cancer, 52,216	
	Esophageal carcinoma, 51	Pancreatic carcinoma, 6608	
	Gastric cancer, 46	Gastric cancer, 5399	
	Pancreatic carcinoma, 32	Lymphoma, 2801	
	Lymphoma, 22	Esophageal carcinoma, 2521	
AE severity			
Serious	1140	96,539	
Nonserious	54	2028	
Reported person			
Physician	327	30,996	
Pharmacist	386	26,808	
Other health professional	341	30,143	
Consumer	75	5298	
Lawyer	3	151	
Unknown	62	5171	

AE = adverse event.

FDA's recommendations. First, exposure assessment was only considered in instances in which oxaliplatin was documented as the "primary suspect." Second, only the most recent version of cases for which follow-up data were available was utilized. Third, instances of the same event, event date, age, gender, and country of origin were identified as duplicate cases.[15] Additionally, reports in which the date of the initial drug intake occurred post-AE occurrence were deemed aberrant and excluded from further analysis. Severe outcomes included life-threatening events or those causing hospitalization, disability, or death. Unexpected oAEs was defined as any significant AE that was not mentioned in the FDA drug prescribing information.

To identify reports of suspected oAEs that deserve special attention in the framework of routine pharmacovigilance activities, irrespective of statistical criteria used to prioritize safety reviews, the designated medical event and important medical event developed by the European Medicines Agency was employed to defined the clinical prioritization of oAEs.[16,17] Additionally, the time to onset of specific oxaliplatin-associated oAEs was also evaluated and defined as the interval between the time of oxaliplatin dosage initiation (START_DT) and the time of oAE onset (EVENT_DT).[18]

Statistical Analysis

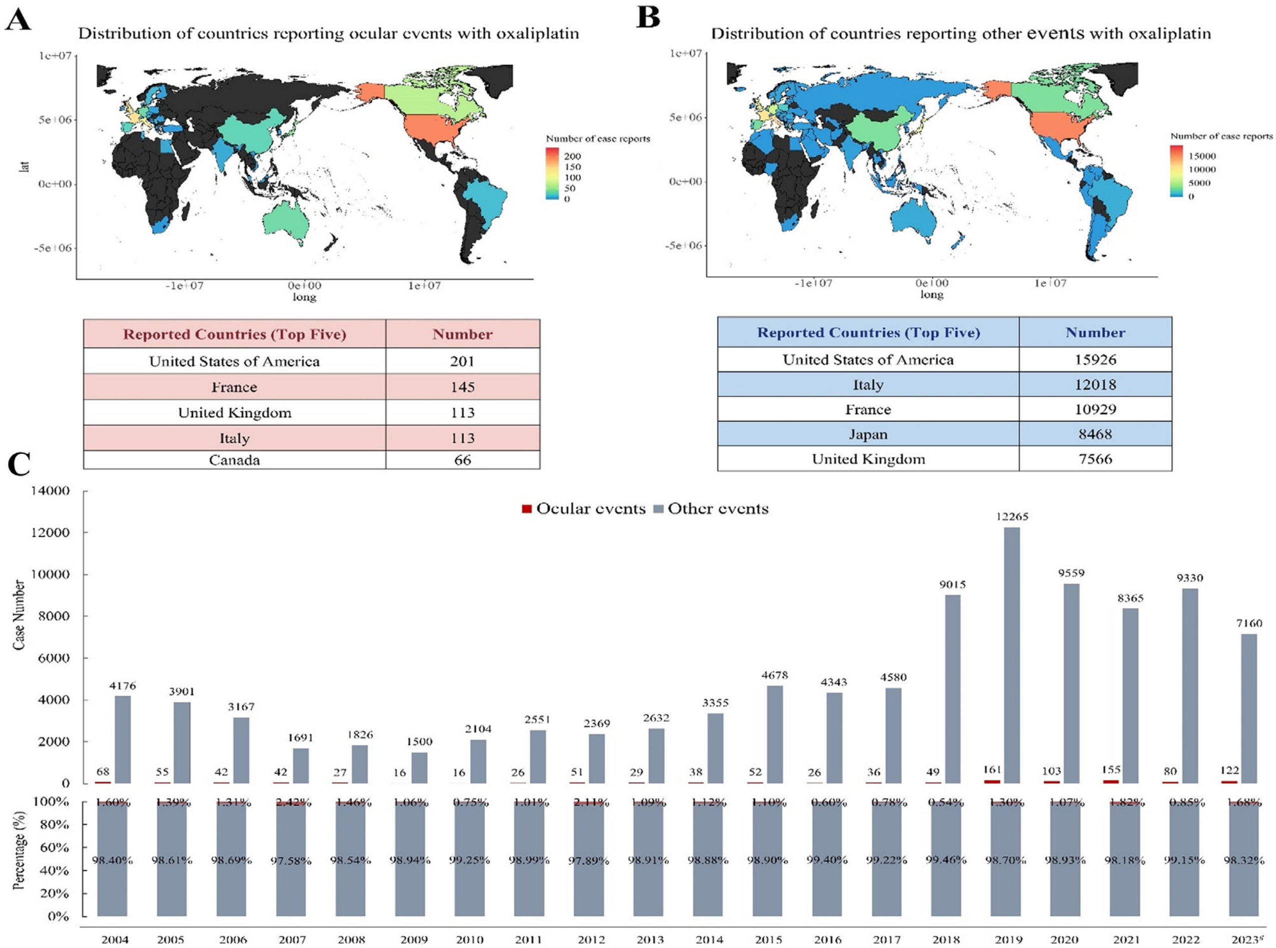
Descriptive statistical analysis was carried out on cases and noncases that had been exposed to oxaliplatin. The baseline characteristics were described in terms of the numbers and percentages for categorical variables.

Disproportionality analysis was performed using the 2 specific indices, the reporting odds ratio (ROR) and Bayesian confidence propagation neural network of information component (IC), to detect potential associations between oxaliplatin and oAEs. Notably, a method known as statistical shrinkage transformation was employed to ensure the robustness of the resulting results, with the relevant calculation formulas and criteria of ROR and IC being provided in Supplemental Tables 1 and 2. In the context of IC, the threshold for a significant signal was defined as the lower limit of the 95% 2-sided CI for IC (IC_{025}) exceeding 0. When $0 < IC_{025} \leq 1.5$, this was classified as a weak signal; between 1.5 and 3.0, it was designated as a middle signal; and above 3.0, it is regarded as a strong signal. As IC-based signals were included in ROR-based ones, ROR was also calculated. For ROR, a significant signal was defined as a lower limit of the 95% CI exceeding 1, with at least 3 cases.[19]

All data processing and statistical analyses were conducted using the statistical software packages SAS 9.4 (SAS Institute, Cary, North Carolina), Microsoft Excel Professional Plus 2013 (Microsoft, Redmond, Washington), and GraphPad Prism 8.0 (GraphPad Software, San Diego, California). Distribution of countries reporting AEs with oxaliplatin analysis were performed by using World_Map 2 tools in Hiplot Pro (https://hiplot.com.cn/), a comprehensive web service for biomedical data analysis and visualization.

Systematic Literature Review

A comprehensive literature review was conducted in PubMed, Embase, and Cochrane Library databases from inception to October 1, 2023 (PROSPERO: CRD42024553009). The search strategy included the keywords ("oxaliplatin") AND ("ocular adverse events" OR "ophthalmic adverse events" OR "ocular toxicity" OR "eye abnormalities" OR "visual field disorders"). MeSH terms ("oxaliplatin" [Mesh] AND "ocular adverse events, ocular toxicity" [Mesh]) were also used in the PubMed search. A report was selected for inclusion if (1) original publication in the English language, (2) available in full text, (3) oAEs were evaluated, (4) cancer patients treated with or in combination with oxaliplatin, and (5) comprehensive description of the case report. Studies were excluded if (1) not related to the defined outcome of interest (oAEs), (2) not related to the defined drug of interest (oxaliplatin), (3) *in vitro* or mechanistic research, and (4) language other than English. The data extraction was conducted by 2 reviewers (W.L. and X.Y.) according to the Preferred Reporting Items for Systematic Reviews and Meta-analyses guidelines.[20] Two reviewers (W.L. and X.Y.) independently searched the literature and then assessed the full text for eligibility after removing duplicates and checking for relevance. One reviewer (H.S.) conducted a snowball search in order to ensure that no relevant articles were missed. Disagreements regarding the extracted data were settled by another re-



Reported Countries (Top Five)	Number
United States of America	201
France	145
United Kingdom	113
Italy	113
Canada	66

Reported Countries (Top Five)	Number
United States of America	15926
Italy	12018
France	10929
Japan	8468
United Kingdom	7566

Figure 2. Reporting country and year of oxaliplatin-related ocular and other events in US Food and Drug Administration Adverse Event Reporting System (FAERS) database. lat = latitude; long = longitude.

viewer (Q.D.). Meanwhile, clinical characteristics including age, gender, indication, chemotherapy regimens, time to onset, clinical characteristics, treatment strategy, and outcome of oAE cases associated with oxaliplatin were also collected. The quality of reporting the cases identified from the literature was evaluated by one reviewer (W.L.) and cross-checked by another (X.Y.) based on the 12 elements required by the International Society for Pharmacoepidemiology and the International Society of Pharmacovigilance for publishing AEs reports,[21] and discrepancies were resolved a third researcher (Q.D.).

Results

Descriptive Analysis of Cohort

A total of 203,402,540 reports were documented in the FAERS database during the surveillance period, which began in January 2004 and concluded in September 2023. Ocular AEs accounted for only a small fraction of the total AEs in all of the oxaliplatin reports, with a case number of 1194, which accounted for 1.20% (1194/99,761) of the overall cases. The specific demographic and clinical details are presented in Table 1. Data on gender were available for 1113 patients, with the proportion of women accounting for 51.57% of the sample. Of the oAE reports associated with oxaliplatin, 45.81% were from patients within the middle age range (18–65 years). Furthermore, the majority of patients (n = 1140, 95.80%) experienced serious outcomes, including hospitalizations (369 cases), required intervention to prevent permanent impairment or damage (14 cases), and life-threatening situations (84 cases), with available follow-up data. In terms of reporting sources, 88.27% (n = 1054) of oAE reports were submitted by the health care professionals. As illustrated in Figures 2A and B, the United States of America, France, United Kingdom, and Italy collectively accounted for the greatest number of oAEs and other event reports (47.91% and 47.11% of the total, respectively). In addition, the number of oAEs per year represented a small but consistently low proportion of the total cases reported for that year (ranging from 0.54% to 2.42%; Figure 2C). Taken as a whole, the proportion of oAEs across various treatment strategies demonstrates that they represent a significant proportion of the overall AE cases associated with oxaliplatin.

Scanning for Oxaliplatin-related oAEs

The High Level Term (HLT) and PT categorization of oAEs was employed in the reports for oxaliplatin, and a disproportionality assessment was conducted through the calculation of the ROR and IC_{025} values, utilizing the comprehensive FAERS database as a reference. In total, 22 significant disproportionality PTs that were simultaneously in accordance with both algorithms are shown in Figure 3. Meanwhile, 9 of the 22 PTs (40.91%) with statistically significant disproportionality signals were categorized as important medical events, and only 2 represented designated medical events, including optic ischemic neuropathy (ROR, 2.77; IC_{025}, 0.48) and toxic optic neuropathy (ROR, 13.22; IC_{025}, 1.50). In the present investigation, 10 PTs of diplopia, blindness transient, ocular toxicity, visual field defect, conjunctival hyperemia, amaurosis fugax, optic ischemic neuropathy, ophthalmoplegia, papilledema,

PTs	Number	IC_{025}	Signal strength of IC	ROR (95% CI)	IME	DME
Retinal vein thrombosis	8	1.02	+	6.36 (3.17-12.77)	N	N
Cataract subcapsular	4	0.06	+	4.23 (1.58-11.33)	N	N
Cataract nuclear	5	0.4	+	5.03 (2.08-12.13)	Y	N
Optic ischaemic neuropathy	12	0.48	+	2.77 (1.57-4.88)	Y	Y
Lacrimation disorder	4	0.75	+	22.90 (8.41-62.38)	N	N
Periorbital oedema	17	0.23	+	1.99 (1.24-3.21)	N	N
Ocular toxicity	28	3.12	+++	22.72 (15.56-33.19)	N	N
Papilloedema	33	1.51	++	4.47 (3.17-6.30)	Y	N
Swelling of eyelid	17	0.42	+	2.31 (1.43-3.72)	N	N
Conjunctival hyperaemia	23	1.23	+	4.01 (2.66-6.05)	N	N
Eyelid oedema	37	0.31	+	1.75 (1.27-2.42)	N	N
Ophthalmoplegia	10	0.48	+	3.04 (1.63-5.66)	Y	N
Blepharospasm	20	0.6	+	2.53 (1.63-3.92)	N	N
Eye movement disorder	22	0.14	+	1.72 (1.13-2.62)	N	N
Eyelid ptosis	58	1.43	+	3.69 (2.85-4.78)	N	N
Blindness cortical	4	0.16	+	4.92 (1.84-13.16)	Y	N
Amaurosis	7	0.84	+	5.91 (2.80-12.45)	Y	N
Toxic optic neuropathy	8	1.5	+	13.22 (6.55-26.67)	Y	Y
Amaurosis fugax	21	2.58	++	15.63 (10.12-24.13)	Y	N
Visual field defect	24	0.15	+	1.71 (1.15-2.55)	N	N
Blindness transient	49	1.53	++	4.10 (3.10-5.44)	Y	N
Diplopia	63	0.22	+	1.51 (1.18-1.94)	N	N

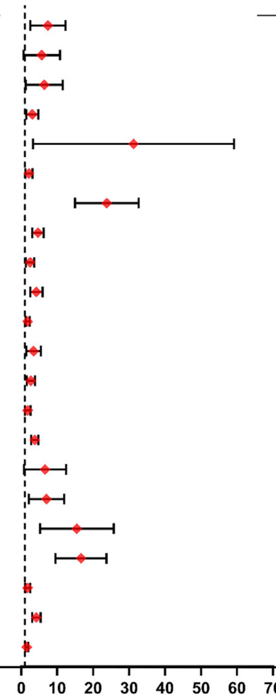


Figure 3. Signal strength of significant ocular adverse events associated with oxaliplatin at the Preferred Terms (PTs) level in US Food and Drug Administration Adverse Event Reporting System database. Values in bold indicates significant signals in 2 algorithms and newly discovered off-label ocular toxicity associated with oxaliplatin. Signals are detected when all the following criteria are met: lower limit of the 95% CI of reported odds ratio (ROR) of >1 and lower limit of the 95% 2-sided CI for information component (IC_{025}) of >0. + = weak; ++ = moderate; +++ = strong; DME = designated medical event; IC = information component; IME = important medical event; N = no; Y = yes.

and amaurosis were identified, which is consistent with observations from previous clinical trials and the oxaliplatin labeling information. Noteworthy, some unexpected oAEs (n = 12) were also detected that were not previously disclosed in the oxaliplatin insert, including eyelid ptosis, eyelid edema, eye movement disorder, blepharospasm, periorbital edema, swelling of the eyelid, ophthalmoplegia, retinal vein thrombosis, cataract nuclear, blindness cortical, cataract subcapsular, and lacrimation disorder. Moreover, in order to reduce data redundancy and confusion and better assess the association between oxaliplatin and oAEs, we summarized the HLT of oAEs involved in oxaliplatin therapy, and the calculation results of the disproportionality analysis are listed in Table 2. As shown in the Supplemental Figure and Table 2, the vision disorders was the most prevalent oAEs associated with oxaliplatin at the High Level Group Term level (accounting for 42.26%), and the eyelid movement disorders (n = 81; ROR, 2.92; IC_{025}, 1.18), ocular nerve and muscle disorders (n = 43; ROR, 1.43; IC_{025}, 0.07), optic disc abnormalities NEC (n = 40; ROR, 3.64; IC_{025}, 1.32), ocular injuries NEC (n = 28; ROR, 22.72; IC_{025}, 3.12), visual field disorders (n = 24; ROR, 1.71; IC_{025}, 0.15), conjunctival infections, irritations, and inflammations (n = 24; ROR, 2.39; IC_{025}, 0.60), visual pathway disorders (n = 21; ROR, 3.06; IC_{025}, 0.86), and optic nerve bleeding and vascular disorders (n = 13; ROR, 2.59; IC_{025}, 0.44) representing the most significant categories of oAEs at the HLT level.

Time to Onset Analysis

More than 50% of cases of oxaliplatin-related oAEs occurred during the first day of initiating oxaliplatin treatments (339/649), with a median onset time of 1 day. This was significantly shorter in the oAE group than in the other events group (1 day [interquartile range, 0–25 days] vs 9 days [interquartile range, 0–39 days]; $P = 0.02$; Figure 4).

Additionally, we further analyzed the statistically significant oAE reports in which the time of onset was reported (n = 249); the onset time of diverse oAEs is detailed in Table 3. Due to the number of reported AEs of cataract subcapsular, cataract nuclear, ocular toxicity, blindness cortical, and toxic optic neuropathy was less than 3, further investigation was not performed.

Outcomes of Patients With Oxaliplatin-related oAEs

As shown in Figure 5, the serious outcomes of the significant disproportionality oAEs associated with oxaliplatin were also evaluated. Compared with other oAEs, there was a higher risk of hospitalization for oAEs of diplopia (29.14%), eyelid ptosis (25.83%), and eye movement disorder (9.93%). Moreover, the oAEs of blindness transient (11.88%), diplopia (7.26%), eyelid edema (10.56%), ocular toxicity (8.91%), and papilledema (9.24%) had a higher risk of other serious outcomes. The FAERS also contains a number of non–drug-related AE signals that may be caused by disease progression or other causes, as well as the causal relationship between serious adverse outcomes and ocular toxicity. Therefore, further analysis of serious outcomes of mortality, life-threatening events, and disability from oxaliplatin-related oAEs was not performed.

Published Case Reports of oAEs Related to Oxaliplatin

The systematic search of the literature databases identified 228 relevant articles. After initial screening, 13 relevant published case reports were included in the qualitative evidence review.[22–34] The detailed process of study screening and exclusion is shown in Figure 1. The quality appraisal of the cases identified from the literature is summarized in Supplemental Table 3. As shown in Table 4 , our literature review identified 20 cases (13 articles) of oAE associated with oxaliplatin. Of these, 8 cases (40%) were male and 12 (60%) were female; 15 (75%) of the patients with oAEs were between 18 and 65 years of age. Furthermore, 15 cases (75%) were colorectal cancers, and 2 (10%) were gastric cancers. The oAEs of 8 cases (40%) occurred in the first cycle

Table 2
Signal strength of ocular reports associated with oxaliplatin at the High Level Term level in US Food and Drug Administration Adverse Event Reporting System database.

HLT	Number	ROR (95%CI)	IC_{025}
Visual impairment and blindness (excluding color blindness)	264	0.71 (0.63–0.80)	−0.67
Visual disorders NEC	190	0.67 (0.58–0.78)	−0.78
Ocular disorders NEC	101	0.40 (033–0.49)	−1.60
Eyelid movement disorders	**81**	**2.92 (2.34–3.63)**	**1.18**
Ocular infections, inflammations, and associated manifestations	76	0.33 (0.26–0.41)	−1.91
Lid, lash, and lacrimal infections, irritations, and inflammations	59	1.09 (0.85–1.41)	−0.25
Lacrimation disorders	53	0.47 (0.36–0.61)	−1.48
Ocular nerve and muscle disorders	**43**	**1.43 (1.06–1.93)**	**0.07**
Optic disc abnormalities NEC	**40**	**3.64 (2.67–4.97)**	**1.32**
Cataract conditions	31	0.33 (0.23–0.46)	−2.09
Retinal bleeding and vascular disorders (excluding retinopathy)	30	1.06 (0.74–1.52)	−0.4
Ocular injuries NEC	**28**	**22.72 (15.56–33.19)**	**3.12**
Pupil disorders	26	0.60 (0.41–0.88)	−1.27
Visual field disorders	**24**	**1.71 (1.15–2.55)**	**0.15**
Conjunctival infections, irritations, and inflammations	**24**	**2.39 (1.60–3.57)**	**0.60**
Visual pathway disorders	**21**	**3.06 (1.99–4.70)**	**0.86**
Ocular sensation disorders	18	0.32 (0.20–0.50)	−2.27
Retinal structural change, deposits, and degeneration	15	0.21 (0.13–0.35)	−2.88
Glaucoma (excluding congenital)	13	0.32 (0.19–0.55)	−2.34
Optic nerve bleeding and vascular disorders	**13**	**2.59 (1.50–4.46)**	**0.44**
Visual color distortions	6	1.75 (0.78–3.89)	−0.44
Ocular bleeding and vascular disorders NEC	5	0.19 (0.08–0.47)	−3.34
Choroid and vitreous structural change, deposit, and degeneration	5	0.19 (0.08–0.46)	−3.35529
Lid, lash, and lacrimal structural disorders	5	0.44 (0.18–1.05)	−2.23
Corneal infections, edemas, and inflammations	4	0.27 (0.10–0.72)	−2.96
Conjunctival and corneal bleeding and vascular disorders	4	0.51 (0.19–1.36)	−2.12
Retinal, choroid, and vitreous infections and inflammations	3	0.10 (0.04–0.43)	−3.94
Corneal disorders NEC	3	0.64 (0.21–1.98)	−1.95

Values in bold indicate significant signals of HLT levels for ocular toxicity associated with oxaliplatin that satisfied 2 algorithms. Signals are detected when all the following criteria are met: lower limit of 95% CI of ROR of >1 and IC_{025} of >0.
HLT = High Level Term; IC_{025} = lower limit of the 95% 2-sided CI for information component; NEC = Not Elsewhere Classified; ROR = reported odds ratio.

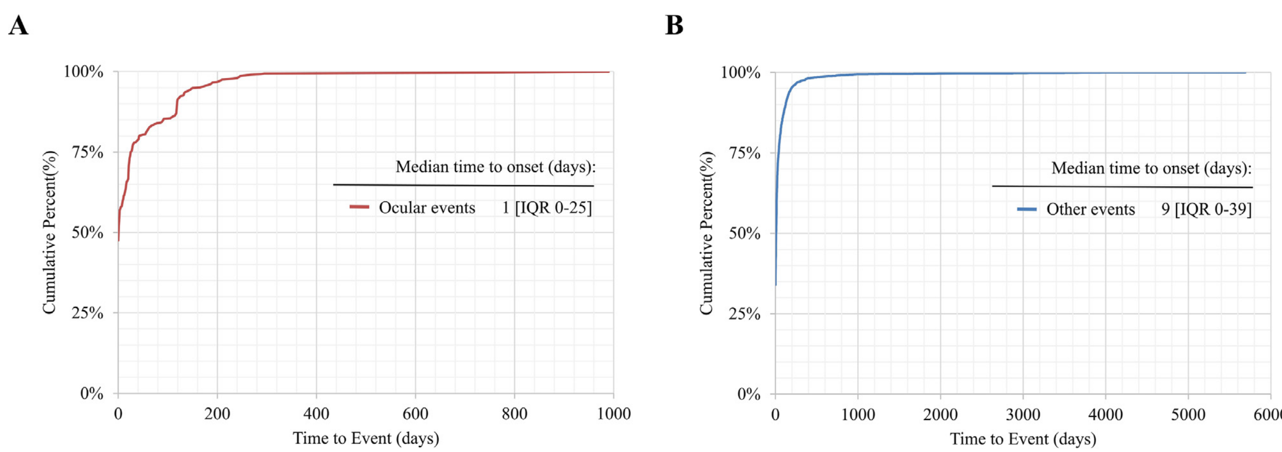


Figure 4. The cumulative distribution curves of the onset time of (A) oxaliplatin-related ocular and (B) other events. IQR = interquartile range.

of oxaliplatin treatment. In the cases included in the study, over half of those who experienced oAEs (n = 11, 55%) suffered loss of vision, with 5 patients displaying blurred vision and another 3 patients experiencing tunnel version. Additionally, there were a number of other oAEs not documented in the package insert (n = 7, 35%), including bilateral blepharoptosis in 4 patients, and papilledema, altered color vision, optic disc swelling in both eyes as oAEs in 3 other patients, respectively. It is noteworthy that, following the discontinuation of oxaliplatin administration or antihistamines and steroid therapy, the majority of patients experienced a complete recovery from the oAEs.

Discussion

Oxaliplatin is the cornerstone of standard chemotherapy regimens in the curative and palliative stages of multiple cancers.[2–6] Similar to other chemotherapy agents, AEs associated with oxaliplatin have been widely reported and studied.[7–9] Ocular AEs, although uncommon, can also affect the quality of survival and cause discomfort in cancer patients on oxaliplatin-based chemotherapy regimens. Clarification of the clinical features of oAEs will help us manage these toxicities as early as possible in order to improve patient tolerability or, ideally, to prevent them from occurring. Actually, there is still a knowledge gap in possible oAEs associated with oxaliplatin. The present study provides a comprehensive summary of the clinical safety spectrum of oAEs associated with oxaliplatin, leveraging a disproportionality analysis and a systematic review of case reports. Moreover, unexpected potential oAEs associated with oxaliplatin were also unraveled.

There are only a few published studies describing oAEs in the setting of oxaliplatin regimens. Previous studies showed that oAEs have mostly been described either in clinical trials or in retrospective case

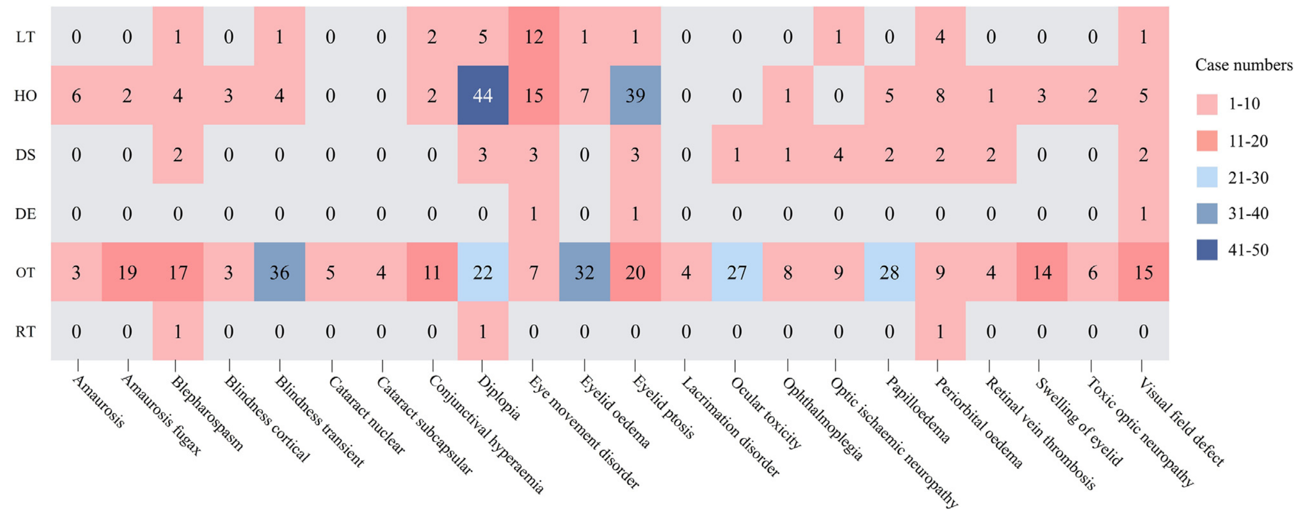


Figure 5. The number of serious outcomes observed in cases of significant ocular adverse events associated with oxaliplatin. DE = death; DS = disability; HO = hospitalization; LT = life-threatening; OT = other serious; RT = required intervention to prevent permanent impairment.

Table 3
Time to onset days of ocular adverse events associated with oxaliplatin.

Ocular AEs	Number	Median (Range)
Diplopia	48*/63†	11 (0–181)
Eyelid ptosis	41/58	8 (0–92)
Eyelid edema	24/37	0 (0–847)
Blindness transient	24/49	0 (0–295)
Papilledema	21/33	2 (0–199)
Conjunctival hyperemia	19/23	1 (0–190)
Eye movement disorder	14/22	0 (0–147)
Periorbital edema	11/17	0 (0–209)
Visual field defect	10/24	0 (0–92)
Blepharospasm	8/20	0 (0–92)
Amaurosis	7/7	0 (0–11)
Swelling of eyelid	5/17	0 (0–0)
Amaurosis fugax	4/21	1 (0–27)
Lacrimation disorder	4/4	0 (0–0)
Optic ischemic neuropathy	3/12	207 (207–232)
Ophthalmoplegia	3/10	134 (21–134)
Retinal vein thrombosis	3/8	0 (0–0)

AE = adverse event.
*Indicates the number of reports that record the specific time of ocular AE occurrence.
† Indicates the number of reports.

reports and small series,[1,11,22–34] but these data only provide a narrow view about these uncommon complications. An randomized controlled trial of oxaliplatin-associated neurotoxicity that used an interview-based questionnaire in patients undergoing oxaliplatin-based chemotherapy with colorectal cancer found that 10% to 15% of patients experienced oAEs, including visual field cuts, ptosis, blurred vision, and eye pain.[35] Furthermore, a retrospective evaluation of 365 patients receiving the FOLFOX (folinic acid + fluorouracil + oxaliplatin) chemotherapy regimen revealed that approximately 38.61% of patients experienced oAEs, and the most prevalent ocular toxic reactions were tearing (52%), decreased visual acuity (58%), blurred vision (34%), dry eyes (29%), and conjunctivitis (15%).[13] It demonstrated that although oAEs related to oxaliplatin are uncommon, they have been documented with a wide variety of symptoms. Furthermore, fluorouracil is also known to cause ocular toxicity,[36,37] but the oAEs of oxaliplatin appear to be greatly underestimated.

In our pharmacovigilance analysis, oAEs are a stable but a less frequently occurring category of AEs associated with oxaliplatin (n = 1194, 1.2%). Over the last 2 decades, 0.54% to 2.42% of all oxaliplatin-related AEs have involved the eye. Meanwhile, approximately 95.48% of patients (n = 1140) experienced serious outcomes based on available follow-up data. The incidence of ocular events as well as serious outcomes were higher than results previously reported in a retrospective real-world study of oxaliplatin safety profile.[1] Such a discrepancy may be attributed to ethnic differences in addition to the limited sample size. Previous case series observed significant corneal changes in patients with peripheral sensory neuropathy caused by oxaliplatin-based chemotherapy treatment.[38] As described in Table 2, our disproportionality analysis showed positive signals for ocular nerve and muscle disorders (n = 43; ROR, 1.43; IC_{025}, 0.07) and optic nerve bleeding and vascular disorders (n = 13; ROR, 2.59; IC_{025}, 0.44) in the HLT levels of oxaliplatin-associated oAEs, which serve to strengthen the conclusion that at least a portion of the oAEs associated with oxaliplatin may be causally related to its neurotoxicity.

A total of 108 oAEs were detected in our current disproportionality analysis, of which 22 oAEs were found to be significantly disproportionate. Among these, approximately 13 were consistent with oAEs disclosed in the oxaliplatin insert, and these findings were further substantiated by the results of our comprehensive literature review, including amaurosis fugax, visual field defect, and papilledema, among others. Noteworthy, as shown in Figure 3, our disproportionality analysis also raise some unexpected safety concerns; the oAEs of eyelid ptosis, eyelid edema, eye movement disorder, blepharospasm, periorbital edema, swelling of eyelid, ophthalmoplegia, retinal vein thrombosis, cataract nuclear, blindness cortical, cataract subcapsular, and lacrimation disorder were first identified about oxaliplatin treatment.

Moreover, no significant disproportionate signal was observed with regard to conjunctivitis as disclosed in the oxaliplatin specification.[10] The underlying reason for these discrepancies may be that this AE is prevalent across all drugs in the FAERS database, resulting in the suppression of the signal score due to the presence of a large number of AE reports associated with multiple drugs. The concept of disproportionality implies that the reporting frequency of drug-specific AEs must be greater (or lower) than that of other AEs. Therefore, the absence of a signal does not imply that oAEs do not exist; rather, it indicates that these oAEs are not disproportional to other AEs.[39]

As described in Table 3, more than 70% of the median time to onset of oxaliplatin-related oAEs occurred within 1 day of treatment, and a recent study showed that corneal changes can be detected up to 2 years after oxaliplatin-based chemotherapy.[40] The median time to onset of diplopia, eyelid ptosis, optic ischemic neuropathy, and ophthalmoplegia in our analysis was 11, 8, 207, and 134 days, respectively, which is

Table 4
Summaries of published case reports of oxaliplatin-related ocular adverse events.

References	Age (y)	Gender	Indication	Chemotherapy regimen	Cycle	Ocular Event	Treatment	Outcome
Tunio et al[22]	52	Male	Colorectal cancer	XELOX	Cycle 1	Loss of vision in the right eye	Discontinuation of oxaliplatin and without any specific treatment	Complete recovery
	26	Male	Colorectal cancer	FOLFIRINOX	Cycle 3, 6	Loss of vision in the right eye	Discontinuation of oxaliplatin at cycle 5 after disease progression	Complete recovery
	54	Female	Colorectal cancer	XELOX	Cycle 2	Blurred vision followed by right eye visual loss	Discontinuation of oxaliplatin	Complete recovery
Noor et al[23]	71	Male	Pancreatic cancer	FOLFIRINOX	Cycle 1, 2	Complete loss of vision in the right eye followed by tunnel vision	Cycle 1: unknown; Cycle 2: discontinuation of oxaliplatin	Fully resolved within 2 d after the first treatment and 5 d after the second treatment
O'Dea et al[24]	45	Female	Colorectal cancer	FOLFOX	Cycle 3	Bilateral visual loss	Discontinuation of oxaliplatin	Ocular symptoms did not recur
	57	Male	Colorectal cancer	FOLFOX	Cycle 1	Tunnel vision		
	52	Male	Hemangiopericytoma	Oxaliplatin monotherapy	Cycle 2	Tunnel vision		
	29	Female	Colorectal cancer	FOLFOX	Cycle 1	Loss of vision in the bottom half of the visual field		
Mesquida et al[25]	52	Female	Colorectal cancer	FOLFOX	Cycle 3	Blurred vision and altered color vision	Chemotherapy discontinuation	Complete resolution
Kubo et al[26]	70	Male	Gastric cancer	SOX	Cycle 1, 2, 3, 4, 5	The visual field of the right eye was impaired	Discontinuation of oxaliplatin	Complete recovery
Patel et al[27]	52	Female	Colorectal cancer	XELOX	Cycle 2	Blurry vision	Without any specific treatment	Visual disturbances did not recur
Painhas et al[28]	54	Female	Colorectal cancer	FOLFOX	*	Papilledema, exclusion and enlarged blind spot in both eyes	Discontinuation of oxaliplatin	Complete resolution
Ah-Thiane et al[29]	51	Female	Colorectal cancer	FOLFOX	Cycle 1	Bilateral visual loss	Discontinuation of oxaliplatin	Complete recovery
	57	Female	Colorectal cancer	FOLFOX	Cycle 3	Right eye visual loss	Discontinuation of oxaliplatin	Complete recovery
Beaumont et al[30]	72	Male	Urothelial cancer	GEMOX	*	Loss of vision and optic disc swelling in both eyes	Discontinuation of oxaliplatin	Complete resolution
Lau and Shibata[31]	76	Female	Gastric cancer	EOX	Cycle 2, 4, 5	Bilateral blepharoptosis and blurred vision in the right eye	Dose reductions of oxaliplatin and the duration of oxaliplatin infusion was extended	All eyelid and visual symptoms ceased to recur
Fanetti et al[32]	54	Female	Colorectal cancer	XELOX	Cycle 3	Bilateral blepharoptosis	Antihistamine and steroid therapy	Symptoms persisted but slowly disappeared
La Verde et al[33]	58	Female	Colorectal cancer	FOLFOX	Cycle 2	Bilateral blepharoptosis	Antihistamine and steroid therapy	After 24 h the ocular symptoms had a complete resolution
	72	Female	Colorectal cancer	Oxaliplatin monotherapy	Cycle 1	Bilateral blepharoptosis	None	Complete resolution
Gilbar and Sorour[34]	50	Male	Colorectal cancer	XELOX + bevacizumab	Cycle 1, 2	Transient blurred and decreased vision in his right eye	Discontinuation of oxaliplatin	*

EOX = epirubicin + capecitabine + oxaliplatin; FOLFIRINOX = folinic acid + fluorouracil + irinotecan + oxaliplatin; FOLFOX = folinic acid + fluorouracil + oxaliplatin; GEMOX = gemcitabine + oxaliplatin; SOX = S-1 + oxaliplatin; XELOX = capecitabine + oxaliplatin.
* Unknown.

consistent with the findings of the previous study. Thus, oxaliplatin-associated ocular toxicity can still occur after the cessation of oxaliplatin treatment. Follow-up should be arranged for further evaluation as needed, as well as close neurologic and ophthalmologic monitoring for early detection and appropriate intervention.

Some studies suggested that oAEs associated with oxaliplatin are rare and often temporary and reversible, usually recovering within 1 to 2 weeks after cessation of the medication.[22–34] None of these patients suffered permanent injury. In our literature review, the majority of patients experienced remission of the associated oAEs simply by discontinuing the medication or not undergoing any treatment. Nevertheless, 2 cases were identified in which a strategy of antihistamine and steroid therapy was employed to manage the patient's oAEs,[32,33] and eventually the ocular symptoms slowly disappeared. In clinical practice, administration of methylcellulose or dexamethasone eye drops can ameliorate ocular symptoms induced by cancer chemotherapy.[41] In short, the methods to treat or prevent oxaliplatin-induced oAEs center around 2 main concepts: dosing strategies and neuromodulatory agents. However, educating the patient and care giver is of utmost importance.

Taken together, oAEs related to oxaliplatin are uncommon and display a wide variety of symptoms, but most of them are transient and reversible.[22] In clinical practice, patients are unlikely to report mild oAEs unless health care practitioners proactively inquire about them. The results of our study suggest that clinicians and pharmacists should pay special attention to the possible oAEs of oxaliplatin because they can occur sporadically, especially in the first day of treatment. A better understanding of the real-world oAE profile of oxaliplatin will contribute to optimal oxaliplatin medication and a better quality of life for cancer patients.

The principal strength of this study is the capacity to identify potential AEs that were not observed in the insert for oxaliplatin. As is common in similar studies conducted using pharmacovigilance databases, this research must be considered in light of several limitations. First, it should be noted that the FAERS database relies upon the voluntary reporting of AEs, which renders it impossible to calculate the incidence or prevalence of these events, and underreporting is to be expected. Second, the reports lack comprehensive clinical data such as previous treatment protocols and disease stage, which impairs the ability to confirm the causal relationship of the oAEs in question. Moreover, the disproportionality analysis did not quantify risk or establish causality, but rather estimated signal strength, which was found to be statistically significant. Therefore, the results of the present study should be considered indicative only, and clinicians and pharmacists should remain vigilant. Consequently, it is imperative that large population-based prospective studies be conducted in order to further elucidate the oAEs associated with oxaliplatin.

Conclusions

The current study uncovered the specific features and filled gaps in our knowledge of oxaliplatin-induced ocular toxicity, leveraging the FAERS database and a systematic review of case reports, which serve as a resource for promoting real-time clinical management and considering the potential risks and benefits of all available therapeutic options for clinicians and patients. Moreover, given the nature of current study, it is imperative to validate our findings in a prospective study and to elucidate the potential mechanisms and risk factors of oxaliplatin-related oAEs in order to inform more effective risk management practices.

Declaration of competing interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Author Contributions

Wensheng Liu and Xuan Ye contributed to collecting, integrating, and visualizing the data and writing the original manuscript. Wensheng Liu, Xuan Ye, Han Shan, Mengmeng Wang, Yingbin Wang, Zihan Guo, Jiyong Liu, and Qiong Du participated in data analysis, interpretation, and manuscript revision. Jiyong Liu and Qiong Du participated in the study design and supervised this research. Wensheng Liu, Xuan Ye, Han Shan, Mengmeng Wang, Yingbin Wang, Zihan Guo, Jiyong Liu, and Qiong Du contributed to the final preparation of this paper and submission.

Data Availability Statement

All data in the article or supplementary material can be freely accessed through the FAERS database. Further reasonable requests can be directed to the corresponding authors.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.clinthera.2024.09.019.

References

1. Yu Z, Huang R, Zhao L, et al. Safety profile of oxaliplatin in 3,687 patients with cancer in China: a post-marketing surveillance study. *Front Oncol*. 2021;11:757196.
2. Adebayo AS, Agbaje K, Adesina SK, Olajubutu O. Colorectal cancer: disease process, current treatment options, and future perspectives. *Pharmaceutics*. 2023;15:2620.
3. Xie Y, Ren Z, Chen H, et al. A novel estrogen-targeted PEGylated liposome co-delivery oxaliplatin and paclitaxel for the treatment of ovarian cancer. *Biomed Pharmacother*. 2023;160:114304.
4. Esim O, Bakirhan NK, Yildirim N, et al. Development, optimization and in vitro evaluation of oxaliplatin loaded nanoparticles in non-small cell lung cancer. *Daru*. 2020;28:673–684.
5. Tempero MA. NCCN guidelines updates: pancreatic cancer. *J Natl Compr Canc Netw*. 2019;17:603–605.
6. Shi M, Yang Z, Lu S, et al. Oxaliplatin plus S-1 with intraperitoneal paclitaxel for the treatment of Chinese advanced gastric cancer with peritoneal metastases. *BMC Cancer*. 2021;21:1344.
7. Rassy E, Le Roy F, Smolenschi C, et al. Rechallenge after oxaliplatin-induced hypersensitivity reactions. *JAMA Oncol*. 2023;9:434–435.
8. Forgie BN, Prakash R, Telleria CM. Revisiting the anti-cancer toxicity of clinically approved platinating derivatives. *Int J Mol Sci*. 2022;23:15410.
9. Hoff PM, Saad ED, Costa F, et al. Literature review and practical aspects on the management of oxaliplatin-associated toxicity. *Clin Colorectal Cancer*. 2012;11:93–100.
10. Ibrahim A, Hirschfeld S, Cohen MH, et al. FDA drug approval summaries: oxaliplatin. *Oncologist*. 2004;9:8–12.
11. Tyler EF, McGhee CNJ, Lawrence B, et al. Corneal nerve changes observed by in vivo confocal microscopy in patients receiving oxaliplatin for colorectal cancer: the COCO study. *J Clin Med*. 2022;11:4770.
12. Vitiello L, Lixi F, Coco G, Giannaccare G. Ocular surface side effects of novel anti-cancer drugs. *Cancers (Basel)*. 2024;16:344.
13. Una E, Alonso P. Ocular toxicities with oxaliplatin in colorectal cancer. *Ann Oncol*. 2014;25(Suppl 2):ii84.
14. Xia S, Gong H, Zhao Y, et al. Tumor lysis syndrome associated with monoclonal antibodies in patients with multiple myeloma: a pharmacovigilance study based on the FAERS database. *Clin Pharmacol Ther*. 2023;114:211–219.
15. Cirmi S, El Abd A, Letinier L, et al. Cardiovascular toxicity of tyrosine kinase inhibitors used in chronic myeloid leukemia: an analysis of the FDA Adverse Event Reporting System Database (FAERS). *Cancers (Basel)*. 2020;12:826.
16. Shu Y, Chen J, Ding Y, Zhang Q. Adverse events with risankizumab in the real world: postmarketing pharmacovigilance assessment of the FDA adverse event reporting system. *Front Immunol*. 2023;14:1169735.

17. Araujo AGS, Borba HHL, Tonin FS, et al. Safety of biologics approved for the treatment of rheumatoid arthritis and other autoimmune diseases: a disproportionality analysis from the FDA Adverse Event Reporting System (FAERS). *BioDrugs*. 2018;32:377–390.
18. Liu W, Du Q, Guo Z, et al. Post-marketing safety surveillance of sacituzumab govitecan: an observational, pharmacovigilance study leveraging FAERS database. *Front Pharmacol*. 2023;14:1283247.
19. Zou SP, Yang HY, Ouyang ML, et al. A disproportionality analysis of adverse events associated to pertuzumab in the FDA Adverse Event Reporting System (FAERS). *BMC Pharmacol Toxicol*. 2023;24:62.
20. Page MJ, Moher D, Bossuyt PM, et al. PRISMA 2020 explanation and elaboration: updated guidance and exemplars for reporting systematic reviews. *BMJ*. 2021;372:n160.
21. Kelly WN, Arellano FM, Barnes J, et al. International Society of Pharmacoepidemiology; International Society of Pharmacovigilance. Guidelines for submitting adverse event reports for publication. *Drug Saf*. 2007;30:367–373.
22. Tunio MA, Phillips K, Baker P. Amaurosis fugax: a rare oxaliplatin-induced ocular toxicity—a report of three cases. *Case Rep Oncol*. 2022;15:133–137.
23. Noor A, Desai A, Singh M. Reversible ocular toxicity of oxaliplatin: a case report. *Cureus*. 2019;11:e4582.
24. O'Dea D, Handy CM, Wexler A. Ocular changes with oxaliplatin. *Clin J Oncol Nurs*. 2006;10:227–229.
25. Mesquida M, Sanchez-Dalmau B, Ortiz-Perez S, et al. Oxaliplatin-related ocular toxicity. *Case Rep Oncol*. 2010;3:423–427.
26. Kubo K, Kimura N, Watanabe R, et al. Oxaliplatin-associated amaurosis fugax. *Case Rep Oncol*. 2021;14:832–837.
27. Patel MA, McDevitt RL, Sassack W, Zalupski MM. Motor "freezing" with oxaliplatin. *J Oncol Pharm Pract*. 2021;27:2014–2017.
28. Painhas T, Amorim M, Soares R, et al. Idiopathic intracranial hypertension and oxaliplatin: a causal association? *Cutan Ocul Toxicol*. 2015;34:237–241.
29. Ah-Thiane L, Raoul JL, Hiret S, et al. Transient vision loss—a rare oxaliplatin-induced ophthalmologic side effect: a report of two cases. *Case Rep Oncol*. 2021;14:483–486.
30. Beaumont W, Sustronck P, Souied EH. A case of oxaliplatin-related toxic optic neuropathy. *J Fr Ophtalmol*. 2021;44:e393–e395.
31. Lau SC, Shibata SI. Blepharoptosis following oxaliplatin administration. *J* Oncol Pharm Pract. 2009;15:255–257.
32. Fanetti G, Ferrari LA, Pietrantonio F, Buzzoni R. Reversible bilateral blepharoptosis following oxaliplatin infusion: a case report and literature review. *Tumori*. 2013;99:e216–e219.
33. La Verde N, Garassino MC, Spinelli G, et al. Reversible palpebral ptosis following oxaliplatin infusion. *Dig Liver Dis*. 2007;39:1041.
34. Gilbar P, Sorour N. Retinal vein thrombosis in a patient with metastatic colon cancer receiving XELOX chemotherapy combined with bevacizumab pre-hepatic resection. *J Oncol Pharm Pract*. 2012;18:152–154.
35. Leonard GD, Wright MA, Quinn MG, et al. Survey of oxaliplatin-associated neurotoxicity using an interview-based questionnaire in patients with metastatic colorectal cancer. *BMC Cancer*. 2005;5:116.
36. Raina AJ, Gilbar PJ, Grewal GD, Holcombe DJ. Optic neuritis induced by 5-fluorouracil chemotherapy: case report and review of the literature. *J Oncol Pharm Pract*. 2020;26:511–516.
37. Jansman FG, Sleijfer DT, de Graaf JC, et al. Management of chemotherapy-induced adverse effects in the treatment of colorectal cancer. *Drug Saf*. 2001;24:353–367.
38. Campagnolo M, Lazzarini D, Fregona I, et al. Corneal confocal microscopy in patients with oxaliplatin-induced peripheral neuropathy. *J Peripher Nerv Syst*. 2013;18:269–271.
39. Sakaeda T, Tamon A, Kadoyama K, Okuno Y. Data mining of the public version of the FDA Adverse Event Reporting System. *Int J Med Sci*. 2013;10:796–803.
40. Chiang JCB, Goldstein D, Trinh T, et al. A cross-sectional study of sub-basal corneal nerve reduction following neurotoxic chemotherapy. *Transl Vis Sci Technol*. 2021;10:24.
41. Schmid KE, Kornek GV, Scheithauer W, Binder S. Update on ocular complications of systemic cancer chemotherapy. *Surv Ophthalmol*. 2006;51:19–40.